User Stories, Use Cases, and Acceptance Criteria

Team members: Owen Mueller, Marcus Rowe

1. Stakeholder Map

Category	Stakeholders
Primary	Vinyl collectors (core users) Project team (Owen and Marcus)
Secondary	Record stores and sellers Data providers / APIs NFC tag suppliers App stores
Hidden	Later student teams who may inherit this project Collectors who cannot or will not adopt the app

---

2. User Stories

ID	Stakeholder Category	Story

US-01	Primary: Vinyl Collector	**As a** vinyl collector with a growing collection, **I want to** keep a digital library of my records, **so that** I can see which records I own by artist, genre, and time period.
US-02	Primary: Vinyl Collector	**As a** vinyl collector, **I want to** see what my collection is worth over time, **so that** I can decide which records to keep or sell.
US-03	Primary: Vinyl Collector	**As a** vinyl collector who plays my records often, **I want to** log a listen without opening the app or typing anything, **so that** I can see how often I play each album.
US-04	Secondary: Data Provider	**As a** provider of record pricing or song data, **I want** the app to stay within my request limits and use my data under my terms, **so that** my service stays available and this project can keep using it.
US-05	Hidden: Later Student Team	**As a** student on a later team who inherits this project, **I want** the code, data model, and setup steps to be clearly documented and runnable, **so that** I can add to the app within days instead of reverse engineering it.

Coverage: three primary stories (US-01 to US-03), one secondary (US-04), and one hidden (US-05).

3. INVEST Self-Check

Each story was checked against **I**ndependent, **N**egotiable, **V**aluable, **E**stimable, **S**mall, and **T**estable.

Story	I	N	V	E	S	T	Notes
US-01	YES	YES	YES	YES	YES	YES	Foundation story that stands alone and states the need, not a screen. Verified by AC-01.1 through AC-01.4.
US-02	YES	YES	YES	YES	YES	YES	Needs records (US-01) and a price source (US-04), but can be built and tested with sample data. Testable by checking that known records and prices give the expected total.
US-03	YES	YES	YES	YES	YES	YES	Reworded to remove the "tapping" interaction, so NFC is one option rather than a requirement. Testable by logging a known number of listens and checking the count.

US-04	YES	YES	YES	YES	YES	YES	Enforcement method (caching, throttling) is left open. Testable by comparing request counts to each provider's published limit.
US-05	YES	YES	YES	YES	YES	YES	Scope is limited to a README, data model description, and setup steps. Testable by having a new developer run the app from the written steps alone.

---

4. Use Cases

UC-01: Filter Collection

Field	Detail
ID	UC-01
Expands	US-01
Name	Filter collection by artist, genre, and decade
Primary actor	Vinyl collector (primary stakeholder)
Secondary actors	None

Preconditions (each can be verified by a tester)

1. The collector is signed in.
2. The collection contains at least 1 record.
3. Every record has an artist, a genre, and a release year.
4. The collection view is reachable from the main page.

Main success flow

Step	Actor action	System response
1	The collector opens the collection view.	The system displays all records sorted by date added, along with a total record count.
2	The collector chooses to filter.	The system offers filter options for artist, genre, and decade.
3	The collector selects Genre = "Rock".	The system displays only records with the genre "Rock" and updates the count.
4	The collector selects Artist = "AC/DC".	The system narrows the list to records that match both Genre = "Rock" and Artist = "AC/DC", and updates the count.

Alternate flow: UC-01-A1 (keyword search instead of preset filters)

At step 2, the collector enters "AC/DC" as a search term instead of choosing a filter option. The system lists every record whose artist, album title, or label contains "AC/DC". The use case then ends at the postcondition.

Exception flow: UC-01-E1 (no records match the filter combination)

At step 3 or 4, no record matches the combined filters. The system displays the message "No records match your filters," keeps all active filters visible, and offers a way to clear them. When the collector clears the filters, the system removes all of them and shows the full collection, as in step 1.

Postcondition

The collector sees a list containing only the records that match every

active filter. No collection data has been changed.

---

5. Acceptance Criteria (Given / When / Then)

UC-01: Filter Collection

AC-01.1: Main success flow (filter by genre and artist)

- **Given** a signed-in collector with 500 records, of which 30 are Rock and 10 of those Rock records are by AC/DC,
- **When** the collector selects Genre = "Rock" and Artist = "AC/DC",
- **Then** exactly 10 records are shown, each with Genre "Rock" and Artist "AC/DC", and the count shows 10.

AC-01.2: Exception flow (no matching records)

- **Given** a signed-in collector with 500 records, none of which are both Rock and by Miles Davis,
- **When** the collector selects Genre = "Rock" and Artist = "Miles Davis",
- **Then** 0 records are shown, the message "No records match your filters" is displayed, and both active filters remain visible.

AC-01.3: Exception flow recovery (clear filters)

- **Given** the no-match state from AC-01.2,
- **When** the collector clears the filters,
- **Then** all 500 records are shown, the count shows 500, and no filters remain active.

AC-01.4: Alternate flow (keyword search)

- **Given** a signed-in collector with 500 records, of which 10 have "AC/DC" in the artist, album title, or label,
- **When** the collector searches for "AC/DC",
- **Then** exactly 10 records are shown, and each contains "AC/DC"

in its artist, album title, or label.